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Abstract

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Title: Developing New Computational Tools to Identify the Most Extreme Outflows in the Universe

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Abstract

Quasars are among the most luminous active galactic nuclei (AGNs) in the universe, powered by massive accretion disks surrounding supermassive black holes at the center of most galaxies. Launching from these accretion disks are energetic outflows of gas and radiation. Among these, quasar extremely high velocity outflows (EHVOs) — defined as outflows exceeding 10% the speed of light — have been confirmed in over 150 cases by our research group, establishing them as a common feature of quasar activity in the early universe rather than rare anomalies. Since kinetic power scales with velocity cubed, EHVOs are suspected to be dominant feedback drivers of early galaxy formation and evolution, including supermassive black hole growth and star formation. However, the fastest EHVOs with speeds greater than 20% the speed of light remain severely underrepresented in existing surveys. Only two have been confirmed to date due to the detection limits of conventional spectral analysis methods.

At these extreme velocities, characteristic ion absorption signatures (CIV, SiIV, NV) are exceptionally blueshifted into the Lyman-alpha (Ly- α) forest region, a dense collection of hydrogen absorption lines from intervening intergalactic gas clouds across the universe that obscures the quasar's intrinsic spectral features. We are developing an automated Python-based detection pipeline to systematically identify EHVOs beyond this $0.2c$ limit. Our approach exploits the physical contrast in variability timescales between EHVO absorption and Ly- α forest lines: by dividing multi-epoch spectral observations from the Sloan Digital Sky Survey (SDSS), the static Ly- α forest is suppressed to baseline, isolating intrinsic variable EHVO features. We have demonstrated this method on a confirmed EHVO candidate, where CIV absorption consistent with $v > 0.2c$ is clearly revealed in spectral ratios and confirmed by our automated absorption detection module. For cases where the ratio method is insufficient, we are integrating SpenderQ, a redshift-invariant autoencoder, to reconstruct the intrinsic quasar continuum without Ly- α forest contamination. Failing to account for EHVOs in this highest velocity range may be causing a significant underestimation of their cosmological influence. This context is directly relevant to interpreting the energetic environments from which gravitational wave signals detected by LISA will emerge.